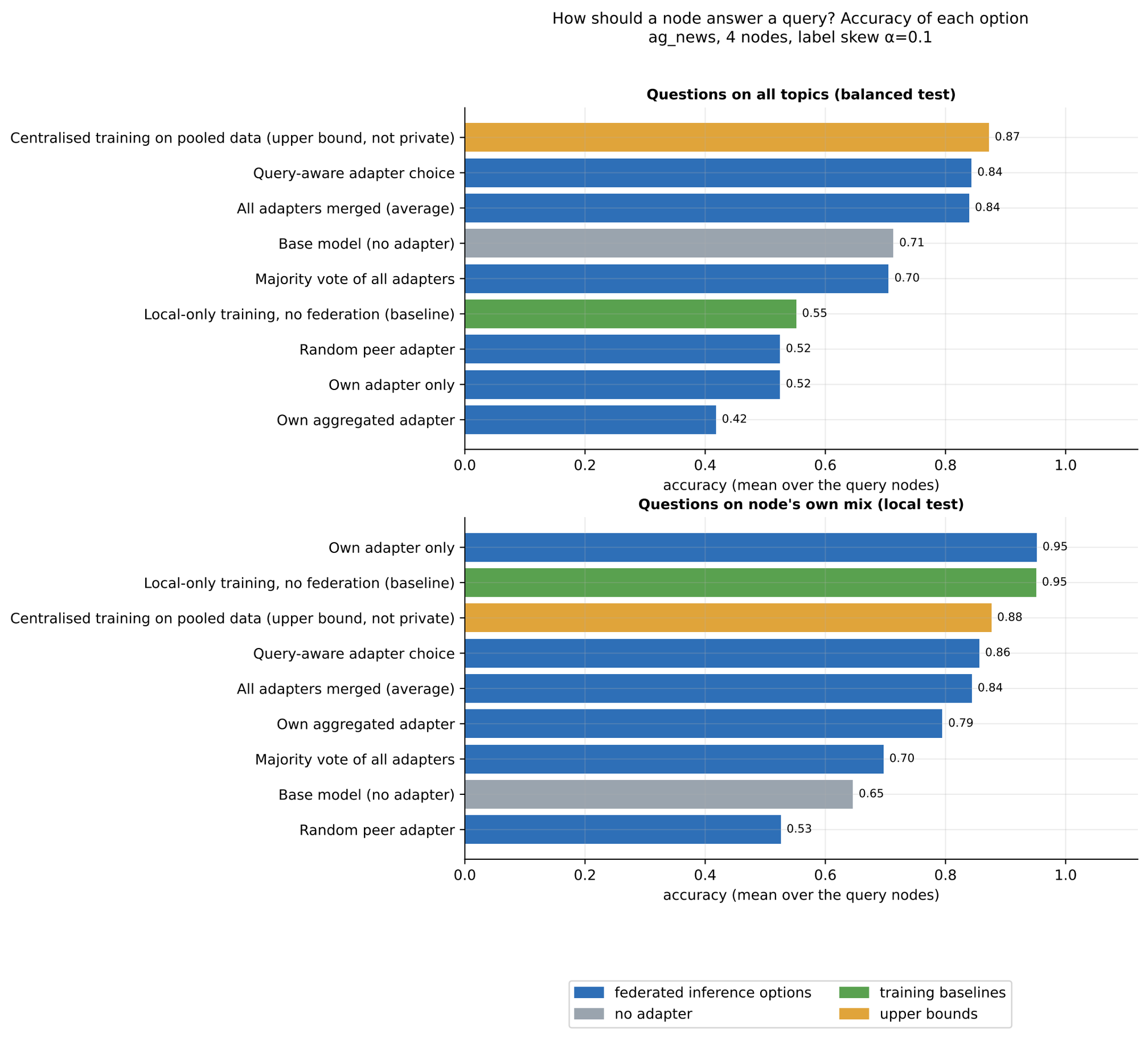
How should a node answer a query? Accuracy of each option
ag_news, 4 nodes, label skew α=0.1
Questions on all topics (balanced test)
Centralised training on pooled data (upper bound, not private)
Query-aware adapter choice
All adapters merged (average)
Base model (no adapter)
Majority vote of all adapters
Local-only training, no federation (baseline)
Random peer adapter
Own adapter only
Own aggregated adapter
0.87
0.84
0.84
0.71
0.70
0.55
0.52
0.52
0.42
0.0
0.2
0.4
0.6
0.8
1.0
accuracy (mean over the query nodes)
Questions on node's own mix (local test)
Own adapter only
Local-only training, no federation (baseline)
Centralised training on pooled data (upper bound, not private)
Query-aware adapter choice
All adapters merged (average)
Own aggregated adapter
Majority vote of all adapters
Base model (no adapter)
Random peer adapter
0.95
0.95
0.88
0.86
0.84
0.79
0.70
0.65
0.53
0.0
0.2
0.4
0.6
0.8
1.0
accuracy (mean over the query nodes)
federated inference options
training baselines
no adapter
upper bounds